Design of Polar Subcodes for Permutation Decoding

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***Abstract*—A novel construction of randomized polar subcodes and subcodes of extended BCH codes is presented. The proposed approach stems from a new heuristic for building polar code frozen set which minimizes the error probability of successive cancellation algorithm applied to permuted noisy codewords. The frame error rate (FER) of the proposed codes under permutation-based decoding is comparable to FER of existing polar subcodes under successive cancellation list or sequential algorithm.**

***Index Terms*—Polar codes, code design, affine group, permutation decoding, successive cancellation list decoding.**

I. Introduction

POLAR codes [1] are a family of error-correcting codes that were shown to attain the capacity of symmetric memoryless channel under successive cancellation (SC) decoding. Due to their low encoding and decoding complexity, polar codes are used in 5G wireless system [2]. The performance of SC for practical blocklengths is quite poor, although successive cancellation list (SCL) [3] decoding allows to obtain near maximum likelihood (ML) performance.

However, polar codes suffer from low minimum distance, which results in poor performance under SCL decoding. In order to enhance SCL decoder, a cyclic redundancy check (CRC) can be appended to an information vector, and then codeword candidates not satisfying this check are eliminated from the list while decoding. More general and effective approach, polar subcodes, is introduced in [4] where dynamic freezing constraints are used instead of CRC. This technique leads to codes having better weight distribution than classical polar codes (and therefore outperforming them under SCL decoding) and in many cases they outperform state-of-art LDPC codes.

Approaches for a fast and efficient implementation of the SCL algorithm were proposed in [5], [6], and [7] and developed in [8], [9], and [10]. Nevertheless, a hardware implementation of the SCL decoder is currently unfeasible when the list size is large. An alternative solution is a permutation decoding algorithm [11] which simultaneously applies instances of SC or SCL (or any other decoder) to permuted versions of the received vector. After that, the inverse permutations are applied to decoders output vectors and the most likely codeword is returned as shown at Fig. 1. Usage of permutations often allows to avoid the situations where a fatal error in a single SC or SCL decoder occurs at an early stage.

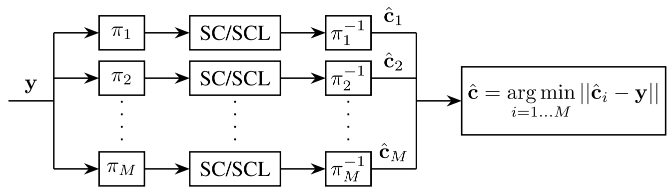

Fig. 1. Scheme of a permutation decoder.

For polar codes it was suggested to use permutations from automorphism group (ones that map any codeword into another codeword), especially affine permutations [12], [13], [14]. Moreover, in [15] it was proposed to use permutations which are not necessary code automorphisms, so the component decoders work with images of a code under corresponding permutations. These codes should have sufficiently low SC error probability in order to enable component decoders to return a correct codeword with high probability.

However, no method of permutation decoding of polar subcodes has been suggested yet. In this work we define the conditions on dynamic freezing constraints and affine permutations that guarantee efficient SC/SCL decoding of permuted polar subcodes. Based on these properties, modified constructions of polar subcodes of extended BCH codes [4] and randomized polar subcodes [16] for permutation decoding are presented. The proposed codes allow efficient decoding of their images under specific affine permutations. Hence, the component decoders (which receive permuted noisy codewords) are able to correct errors of the original decoder. On the other hand, all permutations preserve weight distribution of a code, allowing the component decoders to exploit improved distance properties of polar subcodes. Thus, this paper generalizes the permutation decoding algorithm to polar subcodes which currently can be efficiently decoded only with SCL or sequential [17] decoders.

The rest of the paper is organized as follows. Section II introduces necessary definitions and facts about polar codes and subcodes. In Section III properties of affine permutations that define the structure of permuted polar subcodes are researched. Section IV suggests polar subcode design for permutation decoding. Section V presents simulation results and Section VI summarizes results of the work and describes possible ideas for future researches.

II. Preliminaries

In this article we follow the notations below.

- Vectors and matrices are denoted by bold letters. Their entries are indexed from zero and denoted by lowercase

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letters. For example, x_i is the i-th element of $\mathbf{x}$ and $a_{i,j}$ is the element of $\mathbf{A}$ in the i-th row and the j-th column.
- For a matrix $\mathbf{A}$ we use $\mathbf{A}_{i_1,\ldots,i_r}^{j_1,\ldots,j_s}$ to denote the submatrix of $\mathbf{A}$ spanned by its rows $i_1,\ldots,i_r$ and columns $j_1,\ldots,j_s$. We omit lower or upper subscript if all rows or columns of $\mathbf{A}$ are taken. In particular, $\mathbf{A}_i$ and $\mathbf{A}^j$ denote the i-th row and the j-th column of $\mathbf{A}$ respectively.
- $\mathcal{S}_n$ is the group of all permutations over elements of the set $\{0,1,\ldots,n-1\}$. For $\pi \in \mathcal{S}_n$, $\pi(i)$ denotes action of π on $i \in \{0,1,\ldots,n-1\}$ and $\pi\mathbf{x}$ is permutation of an n-dimensional vector $\mathbf{x}$: $\pi\mathbf{x} = \mathbf{y}$, where $y_i = x_{\pi(i)}$.
- $\mathbb{F}_q$ denotes a finite field of q elements.
- $\mathcal{I}_m$ is the set $\{0,1,\ldots,2^m-1\}$. For $i \in \mathcal{I}_m$ we use $i_0, i_1, \ldots, i_{m-1}$ to denote the digits in the binary expansion (BE) of i, i.e. such integers $i_t \in \{0,1\}$ that $i = \sum_{t=0}^{m-1} i_t \cdot 2^t$. The Hamming weight of i is denoted by $\mathrm{wt}(i) = \sum_{t=0}^{m-1} i_t$.

A. Polar Codes

An $(n = 2^m, k)$ polar code is a binary linear block code with generator matrix obtained by picking from the matrix

$$\mathbf{G}_m = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{\otimes m}$$

k rows with indices of the most reliable virtual bit subchannels under polarization [1]. These indices form the information set $\mathcal{A} \subseteq \mathcal{I}_m$ while $\mathcal{F} = \mathcal{I}_m \setminus \mathcal{A}$ is the set of frozen indices or just frozen set. The codewords have the form $\mathbf{c} = \mathbf{u}\mathbf{G}_m$ where $\mathbf{u}$ is an n-dimensional binary vector having zeros at all the frozen positions.

The subchannel reliabilities can be estimated using Tal-Vardy method [18] or Gaussian approximation [19]. But, regardless the scoring method, the orderings between the reliabilities always coincide with the partial order between corresponding subchannel indices [20], [21]. Consider a weak partial order $\sqsubseteq$ on indices such that for $i, j \in \mathcal{I}_m$ one has $i \sqsubseteq j$ iff $\forall t \in \{0,1,\ldots,m-1\}$ $i_t \le j_t$. The relation $\sqsubseteq$ can be extended to $\preceq$ as follows. If $\mathrm{wt}(i) = \mathrm{wt}(j) = r$, then $i \preceq j$ iff $\forall t \in \{1,2,\ldots,r\}$ $x_t \le y_t$ where $x_1 < x_2 < \ldots < x_r$ and $y_1 < y_2 < \ldots < y_r$ are positions of ones in BEs of i and j respectively. Otherwise $i \preceq j$ iff there is $j^* \in \mathcal{I}_m$ such that $\mathrm{wt}(i) = \mathrm{wt}(j^*)$ and $i \preceq j^* \sqsubseteq j$. Then the statement $i \preceq j$ implies that the i-th subchannel is always less reliable than the j-th one [20]. Thus, we assume that a symbol u_j is unreliable if there is a frozen index with the same Hamming weight as j. Otherwise, if $\mathrm{wt}(j) > \mathrm{wt}(i)$ for any $i \in \mathcal{F}$, u_j is reliable.

SC is a basic decoding algorithm for polar codes [1]. It consists of n phases; let us number them from 0 to $n-1$. At the i-th one it computes a hard decision for the i-th component of the input vector $\mathbf{u} \in \mathbb{F}_2^n$ for $i \notin \mathcal{F}$ or sets $u_i = 0$ otherwise. Algorithm returns $\mathbf{c} = \mathbf{u}\mathbf{G}_m$ as a decoded codeword. But any wrong hard decision immediately leads to a decoding error. In order to improve the performance, SCL algorithm [3] processes at each phase up to L possible prefixes of $\mathbf{u}$ called paths. Instead of making a hard decision for $i \notin \mathcal{F}$ it continues every path with two new paths having $u_i = 0$ or $u_i = 1$ and then eliminates some candidates leaving only L most likely paths alive. In [11] and [19] it was suggested to approximate the error probability of the SC decoder:

$$P_{SC} \approx 1 - \prod_{i \in \mathcal{A}} (1 - \hat{P}_i), \tag{1}$$

where $\hat{P}_i$ is the error probability at the i-th virtual bit subchannel which can be calculated using the method from [19] for Gaussian channel. Analyzing (1), one can conclude that the good SC performance requires the unfrozen symbols to experience low values of $\hat{P}_i$. Or dually, most of the frozen symbols must be unreliable. Although error probability of the SCL algorithm has no analytical formula, let us assume that when P_{SC} gets lower, SCL error probability gets lower as well.

B. Polar Subcodes

Polar subcodes are a generalization of polar codes. A polar subcode of length $n = 2^m$ and dimension k is defined by $(n-k) \times n$ full-rank binary matrix $\mathbf{V}$ having distinct row ends,[1] and $\mathbf{V}$ is called constraint matrix. The codewords have the form $\mathbf{c} = \mathbf{u}\mathbf{G}_m$ where $\mathbf{u} \in \mathbb{F}_2^n$, $\mathbf{u}\mathbf{V}^T = 0$ which is equivalent to

$$u_{r_i} = \sum_{j=0}^{r_i - 1} v_{i,j} u_j, \ 0 \le i < n-k,$$

where r_i is the end of i-th row of $\mathbf{V}$. The condition on u_{r_i} describes dynamic freezing constraint (DFC) if for some $j < r_i$ one has $v_{i,j} = 1$, and the symbol u_{r_i} is dynamic frozen. Otherwise, if $v_{i,j} = 0 \ \forall j < r_i$, we have it static frozen (i.e. $u_{r_i} = 0$). Then frozen set is the set of row ends of $\mathbf{V}$. The decoding algorithms such as SC and SCL can be easily generalized to polar subcodes by setting u_i at phase $i \in \mathcal{F}$ to its dynamically computed value instead of zero.

It was shown in [4] that every binary linear block code of length $n = 2^m$ can be represented as a system of DFCs. Furthermore, when it comes to an extended binary primitive narrow-sense BCH (e-BCH) code, the positions of symbols from its freezing constraints have low Hamming weight and therefore correspond to unreliable bit subchannels. This gives rise to the construction of polar subcodes of e-BCH codes [4]: given design distance d and dimension k, one needs to construct a parent (n, K, d) e-BCH code and then set $K - k$ least reliable unfrozen symbols as static frozen.

Another approach, randomized polar (RP) subcodes, was suggested in [16]. In order to obtain an (n, k) code with parameters f_A and f_B one should construct an $(n, k + f_A)$ polar code called base code. After that one needs to select its k-dimensional subcode by applying f_A random DFCs for unfrozen symbols with maximal indices of minimal weight. More precisely, given the information set $\mathcal{A}$ of a base code, let $r = \min_{i \in \mathcal{A}} \mathrm{wt}(i)$, $\mathcal{A}_w = \{i \in \mathcal{A} \mid \mathrm{wt}(i) = w\}$ and let $\{i_1, i_2, \ldots, i_{f_A}\}$ be the set of f_A maximal elements in $\mathcal{A}_r$. If $|\mathcal{A}_r| < f_A$, maximal elements from $\mathcal{A}_{r+1}, \mathcal{A}_{r+2}, \ldots$ are added. Then for every $l = 1, \ldots, f_A$ one needs to

[1] For a binary vector $\mathbf{v}$ let us define $\mathrm{end}(\mathbf{v}) = \max\{i \mid v_i = 1\}$, so that $\mathbf{v}$ ends at the $\mathrm{end}(\mathbf{v})$-th position.

select random binary equiprobable values $(a_{l,t})_{t=0,\dots,i_l-1}$, so the subcode consists of all vectors $\mathbf{u}\mathbf{G}_m$ of a base code such that for every $l = 1,\dots,f_A$ one has $u_{i_l} = \sum_{t=0}^{i_l-1} a_{l,t}u_t$: these DFCs are called type-A DFCs. It is equivalent to adding to the constraint matrix of a base code rows $(a_{l,0}, a_{l,1},\dots,a_{l,i_l-1}, 1, 0,\dots,0) \in \mathbb{F}_2^n$ for $l = 1,\dots,f_A$. After that f_B most reliable frozen symbols in a base code are to be set dynamic frozen the same way, and underlying DFCs are called type-B DFCs. The purpose of type-A DFCs is to eliminate low-weight codewords from a base code, while type-B DFCs improve performance of the SCL algorithm.

C. Affine Permutations

Polar codes can be viewed in terms of boolean functions. For $\alpha \in \mathcal{I}_m$ let us define a monomial of m variables $g_\alpha(x_0, x_1,\dots,x_{m-1}) = x_0^{\alpha_0}x_1^{\alpha_1}\dots x_{m-1}^{\alpha_{m-1}} = \prod_{i:\alpha_i=1} x_i$. These monomials form the set $\mathcal{M}_m = \{g_\alpha \mid \alpha \in \mathcal{I}_m\}$ with multiplication operation which is logical conjunction. Note that elements of $\mathcal{M}_m$ are a basis of $\mathcal{B}_m$, the linear space over $\mathbb{F}_2$ of boolean functions of m variables where sum is logical xor. There is the isomorphism ev between $\mathcal{B}_m$ and $\mathbb{F}_2^n, n = 2^m$ which maps every $f \in \mathcal{B}_m$ to its evaluation $\mathbf{z} \in \mathbb{F}_2^n$ such that $z_i = f(i_0, i_1,\dots,i_{m-1})\ \forall i \in \mathcal{I}_m$, where digits of the BE are interpreted as booleans. Moreover, it can be seen [20] that the i-th row of $\mathbf{G}_m^T$ equals $\mathrm{ev}(g_i)$.

This formalism is useful for working with $\mathrm{GA}(m)$, group of affine permutations over $\mathcal{I}_m$. Each of them is defined by a non-singular matrix $\mathbf{A} \in \mathbb{F}_2^{m\times m}$ (factor) and a vector $\mathbf{b} \in \mathbb{F}_2^m$ (shift) and acts as $i \mapsto j$ where for $\mathbf{x} = (i_0, i_1,\dots,i_{m-1})$ and $\mathbf{x}' = (j_0, j_1,\dots,j_{m-1})$ one has

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{b}.$$

Here $\mathbf{x}$ and $\mathbf{x}'$ are column vectors from $\mathbb{F}_2^m$, i.e. digits in the BEs of i and j are interpreted as elements of $\mathbb{F}_2$. For $\pi \in \mathrm{GA}(m)$ the notation $\pi : (\mathbf{A}, \mathbf{b})$ defines π as an affine permutation with the factor $\mathbf{A}$ and the shift $\mathbf{b}$. Action of $\pi \in \mathrm{GA}(m)$ on $\mathbb{F}_2^n$ transforms boolean functions corresponding to elements of $\mathbb{F}_2^n$ under the isomorphism ev. Namely, let $\pi : (\mathbf{A}, \mathbf{b})$ and $F_\pi : \mathcal{B}_m \to \mathcal{B}_m$ be the substitution over variables $x_i \mapsto \sum_{j=0}^{m-1} a_{i,j}x_j + b_i$. In other words, F_π acts on the monomial $x_{i_1}x_{i_2}\dots x_{i_r}$ as $\prod_{p=1}^{r}\left(\sum_{j=0}^{m-1} a_{i_p,j}x_j + b_{i_p}\right)$ and on polynomials F_π is defined linearly. Then for every $f \in \mathcal{B}_m$ one has

$$\pi\,\mathrm{ev}(f) = \mathrm{ev}(F_\pi(f)). \tag{2}$$

There are some subgroups of $\mathrm{GA}(m)$ that are particulary useful for the proposed code construction. Firstly, $\mathrm{LTA}(m)$ is the group of affine permutations with lower-triangular factor matrix. Secondly, $\mathrm{BDA}(s_1, s_2,\dots,s_t) \triangleq \mathrm{BDA}(\mathbf{s}) \subseteq \mathrm{GA}(m)$ is the group of affine permutations with block-diagonal factor matrix, where $\mathbf{s} = [s_1, s_2,\dots,s_t]$ is the list of the block sizes: it consists of positive integers s_i such that $\sum_{i=1}^{t} s_i = m$. So, the factor matrix has the form $\mathrm{diag}(\mathbf{B}_1, \mathbf{B}_2,\dots,\mathbf{B}_t)$ where $\mathbf{B}_i$ is a non-singular $(s_i \times s_i)$-matrix. Examples of factor matrices for LTA and BDA permutations are given in Fig. 2.

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ a_{1,0} & 1 & 0 & 0 & 0 & 0 \\ a_{2,0} & a_{2,1} & 1 & 0 & 0 & 0 \\ a_{3,0} & a_{3,1} & a_{3,2} & 1 & 0 & 0 \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & 1 & 0 \\ a_{5,0} & a_{5,1} & a_{5,2} & a_{5,3} & a_{5,4} & 1 \end{pmatrix}$$

(a) LTA(6).

$$\left(\begin{array}{c|cc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c|}{\multirow{2}{*}{$\mathbf{B}_2$}} & 0 & 0 & 0 \\ 0 & & & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & \mathbf{B}_3 & \\ 0 & 0 & 0 & & & \end{array}\right)$$

(b) $\mathrm{BDA}(1, 2, 3)$.

Fig. 2. Factor matrices for LTA and BDA permutations.

D. Permutation Decoding

We propose to apply SC/SCL decoder instances to vectors that are permuted noisy codewords. Consider a polar subcode $\mathcal{C}$ of length $n = 2^m$ with constraint matrix $\mathbf{V}$, $\pi \in \mathcal{S}_n$ and $\mathcal{C}' = \{\pi\mathbf{c} \mid \mathbf{c} \in \mathcal{C}\}$. It was shown in [22] that constraint matrix $\mathbf{V}'$ of $\mathcal{C}'$ satisfies

$$\mathbf{V}' \cong \mathbf{V}(\mathbf{G}_m\mathbf{P}^{-1}\mathbf{G}_m)^T = \mathbf{V}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T, \tag{3}$$

where $\mathbf{P} \in \mathbb{F}_2^{n\times n}$ is the permutation matrix of π (i.e. $p_{i,j} = 1$ iff $i = \pi(j)$) and $\cong$ means equality up to an invertible row transformation.

Example 1: Let $\mathcal{C}$ be the code with frozen set $\{0, 1, 2, 4\}$,

$$\mathbf{V} = \begin{pmatrix} 1&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&1&1&0&0&0 \end{pmatrix} \text{ and } \mathbf{P} = \begin{pmatrix} 0&0&0&0&0&0&0&1 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&0&1&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&0&1&0&0&0 \\ 1&0&0&0&0&0&0&0 \\ 0&0&0&0&0&1&0&0 \\ 0&1&0&0&0&0&0&0 \end{pmatrix}$$

what leads to

$$\mathbf{V}' = \begin{pmatrix} 1&0&0&0&0&0&0&0 \\ 1&0&0&0&1&0&0&0 \\ 0&1&1&0&0&0&0&0 \\ 1&1&0&0&0&1&1&0 \end{pmatrix}.$$

This defines the frozen set $\{0, 2, 4, 6\}$ of $\mathcal{C}'$. According to (1), in case of the AWGN channel with $E_b/N_0 = 1.5$ dB for the original code $P_{SC} \approx 0.177$ and for the permuted one $P_{SC} \approx 0.57$. Thus, the permuted code has much worse performance under SC and SCL decoding.

Permutation decoding algorithm is based on a predefined list of permutations $\pi_1, \pi_2,\dots,\pi_M \in \mathcal{S}_n$. For $i = 1,\dots,M$ let $\mathcal{C}_i = \{\pi_i\mathbf{c} \mid \mathbf{c} \in \mathcal{C}\}$ and dec_i be a decoder of $\mathcal{C}_i$ (usually SC or SCL). Given a received vector $\mathbf{y}$, the algorithm obtains the codeword candidates $\hat{\mathbf{c}}_i = \pi_i^{-1}\mathrm{dec}_i(\pi_i\mathbf{y})$ and returns the most likely one with respect to $\mathbf{y}$.

It is important to properly select these permutations, especially in case of small list size. The concept of equivalent permutations (i.e. permutations that does not affect SC/SCL decoder to return distinct codewords) is introduced in [13]. This relation splits the group $\mathrm{GA}(m)$ into equivalence classes, so that application of the decoder to all permutations of a noisy vector from the same class results in the same decoder output. To check that permutations with factor matrices $\mathbf{A}_1$ and $\mathbf{A}_2$ are equivalent it is enough to check that the permutation with the factor matrix $\mathbf{A}_1\mathbf{A}_2^{-1}$ is not SC-invariant, i.e. not equivalent to the identity permutation. The complete definition of SC-invariant subgroup of affine permutations is given in [23].

III. Properties of Affine Permutations

Before we move to the proposed code design algorithms, let us explore how an affine permutation modifies constraint system of a polar subcode. Our goal is to ensure that for a fixed code $\mathcal{C}$ and a permutation π the SC error probability for the permuted code (i.e. the image of $\mathcal{C}$ under π) is sufficiently low. This allows to use π in the permutation decoder of $\mathcal{C}$. In other words, the frozen set of the permuted code $\mathcal{C}'$ is required to provide sufficiently low value of (1). Therefore, it is important to investigate maximal positions of non-zero elements in the rows of the constraint matrix of $\mathcal{C}'$ which represent the frozen indices. This section is dedicated to the properties of affine permutations that describe transformation of freezing constraints.

A. Affine Substitution in a Monomial Function

Definition 1: Let $f \in \mathcal{B}_m,\ g \in \mathcal{M}_m$ and $I \subseteq \mathcal{M}_m$ such that

$$f = \sum_{\mu \in I} \mu.$$

We say $f \vdash g$ iff $g \in I$, i.e. the decomposition of f with respect to $\mathcal{M}_m$ contains g.

Note that I in Definition 1 is defined uniquely since $\mathcal{M}_m$ is a basis of $\mathcal{B}_m$ which is a linear space over a binary field.

Let us study how the transformation (3) affects a row $\mathbf{v}$ of a constraint matrix $\mathbf{V}$ where $\mathbf{P}$ is matrix of some affine permutation π. The vector $\mathbf{v}$ can be decomposed using the standard basis of $\mathbb{F}_2^n$: $\mathbf{v} = \sum_{i=0}^{n-1} v_i \mathbf{e}_i$, where $\mathbf{e}_i = (0,\ldots,0,\underset{(i)}{1},0,\ldots,0) \in \mathbb{F}_2^n$. In can be seen that (3) transforms the basis elements in the following way:

$$\begin{aligned}\mathbf{e}_i\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T &= \operatorname{ev}(g_i)\mathbf{P}\mathbf{G}_m^T = (\pi\operatorname{ev}(g_i))\mathbf{G}_m^T\\ &= \operatorname{ev}(F_\pi(g_i))\mathbf{G}_m^T = \operatorname{ev}\left(\sum_{j:F_\pi(g_i)\vdash g_j} g_j\right)\mathbf{G}_m^T\\ &= \sum_{j:F_\pi(g_i)\vdash g_j}\operatorname{ev}(g_j)\mathbf{G}_m^T = \sum_{j:F_\pi(g_i)\vdash g_j}\mathbf{e}_j. \quad (4)\end{aligned}$$

Therefore, we need to explore how the set of the indices j in the last statement in (4) depends on i and π.

Definition 2: Let $\pi \in \mathrm{GA}(m),\ i \in \mathcal{I}_m$. Then $\mathcal{G}_\pi(i)$ is the set of all $j \in \mathcal{I}_m$ such that $F_\pi(g_i) \vdash g_j$.

Remark 1: According to (4), $\mathcal{G}_\pi(i)$ is the set of all positions of ones of the vector $\mathbf{e}_i\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$, where $\mathbf{P}$ is matrix of π.

Example 2: Rows of constraint matrix of the permuted code from the Example 1 can be easily obtained using monomial transformation. In fact, $\mathbf{P}$ defines the affine permutation

$$\pi : \mathbf{A} = \begin{pmatrix}0&0&1\\1&1&0\\0&1&0\end{pmatrix},\ \mathbf{b} = \begin{pmatrix}1\\0\\1\end{pmatrix}.$$

Thus, for the equation $u_0 = 0$ (corresponding to $\mathbf{V}_0 = \mathbf{e}_0$) one has $F_\pi(g_0) = g_0$ and hence $\mathcal{G}_\pi(0) = \{0\}$, for $u_1 = 0$: $F_\pi(g_1) = g_0 + g_4$, so $\mathcal{G}_\pi(1) = \{0,4\}$, for $u_2 = 0$: $F_\pi(g_2) = g_1 + g_2$, so $\mathcal{G}_\pi(2) = \{1,2\}$. The last row of $\mathbf{V}$ is $\mathbf{e}_3 + \mathbf{e}_4$, so in order to transform its equation $u_3 + u_4 = 0$ one needs to take the symmetric difference of the sets $\mathcal{G}_\pi(3) = \{1,2,5,6\}$ and $\mathcal{G}_\pi(4) = \{0,2\}$ getting $\{0,1,5,6\}$ which corresponds to the decomposition $F_\pi(g_3 + g_4) = F_\pi(g_3) + F_\pi(g_4) = g_0 + g_1 + g_5 + g_6$.

The following facts investigate the structure of the set $\mathcal{G}_\pi(i)$, i.e. which monomials are present in the decomposition with respect to $\mathcal{M}_m$ after an "affine" transform F_π of g_i. To avoid collisions, in what follows we sometimes use greek letters to denote elements of $\mathcal{I}_m$.

Lemma 1: Let $\mathbf{A} \in \mathbb{F}_2^{k\times m}$, $\mathbf{b} \in \mathbb{F}_2^k$,

$$f(x_0,x_1,\ldots,x_{m-1}) = \prod_{i=0}^{k-1}\left(\sum_{j=0}^{m-1} a_{i,j}x_j + b_i\right)$$

and $f \vdash g_\alpha$ such that $\mathrm{wt}(\alpha) = r$. Then $\operatorname{rank}\mathbf{A} \geq r$.

Proof: The proof is given in Appendix. □

The converse is true when $\mathbf{A}$ is a square matrix and r is its dimension.

Lemma 2: Let $\mathbf{A} \in \mathbb{F}_2^{m\times m}$, $\mathbf{b} \in \mathbb{F}_2^m$,

$$f(x_0,x_1,\ldots,x_{m-1}) = \prod_{i=0}^{m-1}\left(\sum_{j=0}^{m-1} a_{i,j}x_j + b_i\right) \quad (5)$$

and $\mu(x_0,x_1,\ldots,x_{m-1}) = \prod_{j=0}^{m-1} x_j$. Then $f \vdash \mu$ iff $\mathbf{A}$ is non-singular.

Proof: There is a bijection between the elements of $\mathcal{S}_m$ and the ways how to pick summands from the multipliers in (5), one from each one, in order to obtain $\mu(x_0,x_1,\ldots,x_{m-1})$. Indeed, let i be mapped to j if $a_{i,j}x_j$ is chosen from the i-th multiplier. Note, that there is no way to pick any variable twice or pick b_i because there are only m multipliers and all the m variables must be present in the resulting monomial. Thus, in the decomposition of f with respect to $\mathcal{M}_m$ the coefficient of μ is

$$\sum_{\sigma\in\mathcal{S}_m} a_{0,\sigma(0)}a_{1,\sigma(1)}\ldots a_{m-1,\sigma(m-1)} = \det\mathbf{A}.$$

□

Theorem 1: Let $\pi : (\mathbf{A},\mathbf{b}) \in \mathrm{GA}(m),\ \alpha,\beta \in \mathcal{I}_m,\ \mathrm{wt}(\alpha) = \mathrm{wt}(\beta) = r$. Then $\beta \in \mathcal{G}_\pi(\alpha)$ iff the matrix $\mathbf{A}_{i_1,\ldots,i_r}^{j_1,\ldots,j_r}$ is non-singular where $i_1,i_2,\ldots,i_r$ and $j_1,j_2,\ldots,j_r$ are positions of ones in binary expansions of α and β respectively.

Proof: Since $g_\beta(x_0,x_1,\ldots,x_{m-1}) = x_{j_1}x_{j_2}\ldots x_{j_r}$ is present in the decomposition of $\prod_{p=1}^{r}\left(\sum_{j=0}^{m-1} a_{i_p,j}x_j + b_{i_p}\right)$, it is present in the decomposition of the same expression where all the "odd" variables are eliminated, namely $\prod_{p=1}^{r}\left(\sum_{q=1}^{r} a_{i_p,j_q}x_{j_q} + b_{i_p}\right)$. By Lemma 2 this fact is equivalent to $\det\mathbf{A}_{i_1,\ldots,i_r}^{j_1,\ldots,j_r} \neq 0$. □

Our focus is to describe maximal element of the set $\mathcal{G}_\pi(\alpha)$ since maximal elements in rows of a constraint matrix define frozen set. Theorem 1 states which indices β such that $\mathrm{wt}(\alpha) = \mathrm{wt}(\beta)$ are present in $\mathcal{G}_\pi(\alpha)$. Next, we consider indices of other Hamming weight than α and conclude that they can not be maximal.

Proposition 1: Let $\alpha \in \mathcal{I}_m,\ \pi \in \mathrm{GA}(m),\ \alpha' \in \mathcal{G}_\pi(\alpha)$. Then $\mathrm{wt}(\alpha') \leq \mathrm{wt}(\alpha)$.

Proof: Let $\pi : (\mathbf{A},\mathbf{b})$. It is enough to notice that $F_\pi(g_\alpha) = \prod_{i:\alpha_i=1}\left(\sum_{j=0}^{m-1} a_{i,j}x_j + b_i\right)$ consists of $\mathrm{wt}(\alpha)$ linear multipliers. □

Definition 3: Let $\alpha,\beta \in \mathcal{I}_m$. Then $\alpha \sqsubset \beta$ (α is strictly covered by β) iff $\alpha \sqsubseteq \beta$ and $\alpha \neq \beta$.

Note that $\alpha \sqsubset \beta$ implies $\alpha < \beta$. The following lemma establishes that every element of $\mathcal{G}_\pi(\alpha)$ is covered by some element with the same Hamming weight as α.

Lemma 3: Let $\alpha \in \mathcal{I}_m,\ \pi \in \mathrm{GA}(m),\ \beta \in \mathcal{G}_\pi(\alpha)$. Then there exists $\beta' \in \mathcal{G}_\pi(\alpha)$ such that $\beta \sqsubseteq \beta'$ and $\mathrm{wt}(\beta') = \mathrm{wt}(\alpha)$.

Proof: Consider $k = \mathrm{wt}(\alpha)$, $r = \mathrm{wt}(\beta)$ and $r < k$, otherwise $r = k$ by Proposition 1, so $\beta' = \beta$ fulfills the lemma.

Without loss of generality $\alpha_i = 1$ for $0 \le i < k$ and $\beta_i = 1$ for $0 \le i < r$. Let $\pi : (\mathbf{A}, \mathbf{b})$, $\mathbf{S} = \mathbf{A}_{0,\dots,k-1}$ and $\mathbf{U} = \mathbf{A}_{0,\dots,k-1}^{0,\dots,r-1} = \mathbf{S}^{0,\dots,r-1}$. Since $\beta \in \mathcal{G}_\pi(\alpha)$, $\prod_{i=0}^{k-1}\left(\sum_{j=0}^{r-1} a_{i,j}x_j + b_i\right) \vdash x_0x_1\dots x_{r-1}$ and by Lemma 1 $\operatorname{rank}\mathbf{U} = r$, hence columns of $\mathbf{U}$ are linearly independent. But all rows of $\mathbf{A}$ are linearly independent by definition, so the linear space spanned by columns of $\mathbf{S}$ is k-dimensional. The set of columns of $\mathbf{U}$ can be extended with some extra $k - r$ columns $\mathbf{S}^{j_r}, \mathbf{S}^{j_{r+1}}, \dots, \mathbf{S}^{j_{k-1}}$ to form the basis, i.e. linearly independent set of k vectors. Consider β' such that $\beta'_0 = \dots = \beta'_{r-1} = \beta'_{j_r} = \dots = \beta'_{j_{k-1}} = 1$ and $\beta'_i = 0$ for other indices i. Then the result follows from Theorem 1. □

Remark 2: Combining Lemma 3 and Proposition 1 one can obtain that $\mathrm{wt}(\max \mathcal{G}_\pi(\alpha)) = \mathrm{wt}(\alpha)$.

Finally, we observe that if $\alpha \sqsubset \beta$, then every affine permutation extends the relation "$\sqsubset$" to pairs of elements from $\mathcal{G}_\pi(\alpha)$ and $\mathcal{G}_\pi(\beta)$.

Lemma 4: Let $\alpha, \beta \in \mathcal{I}_m,\ \alpha \sqsubset \beta,\ \pi \in \mathrm{GA}(m)$. Then for all $\alpha' \in \mathcal{G}_\pi(\alpha)$ there exists $\beta' \in \mathcal{G}_\pi(\beta)$ such that $\alpha' \sqsubset \beta'$.

Proof: Let $r = \mathrm{wt}(\alpha)$, $k = \mathrm{wt}(\beta)$, so $r < k$. It is sufficient to investigate such α' that $\mathrm{wt}(\alpha') = r$, otherwise α' is strictly covered by some $\alpha'' \in \mathcal{G}_\pi(\alpha)$ with $\mathrm{wt}(\alpha'') = r$. Let $i_1, i_2, \dots, i_r$ and $j_1, j_2, \dots, j_r$ be positions of ones in binary expansion of α and α' respectively; $i_{r+1}, \dots, i_k$ are positions of ones in β corresponding to zeros in α. By Theorem 1 $\det \mathbf{A}_{i_1,\dots,i_r}^{j_1,\dots,j_r} \ne 0$, so the columns $j_1, j_2, \dots, j_r$ of the matrix $\mathbf{A}_{i_1,\dots,i_r}$ are linearly independent, and therefore for the matrix $\mathbf{S} = \mathbf{A}_{i_1,\dots,i_k}$ its columns $\mathbf{S}^{j_1}, \mathbf{S}^{j_2}, \dots, \mathbf{S}^{j_r}$ are linearly independent as well. Since $\mathbf{S}$ is a full-rank $(k \times m)$-matrix, there exist indices $j_{r+1}, \dots, j_k$ such that the columns $\mathbf{S}^{j_1}, \mathbf{S}^{j_2}, \dots, \mathbf{S}^{j_k}$ form a basis of k-dimensional vector space. For β': $\beta'_{j_1} = \beta'_{j_2} = \dots = \beta'_{j_k} = 1$ one has $\beta' \in \mathcal{G}_\pi(\beta)$ by Theorem 1 and $\alpha' \sqsubset \beta'$. □

B. Transformation of a Constraint System Under Affine Permutations

Let us formulate requirements for an affine permutation to enable efficient error correction in a permutation-based decoder. On the first hand, there is an affine subgroup of permutations which are always SC-invariant [13]. This group is formed by permutations whose factor matrix $\mathbf{A}$ has $a_{i,j} = 0$ whenever $0 < i < j$ or $i = 0, j > 1$ [12], [13]. Thus, one should use permutations outside of this group.

On the other hand, an arbitrary affine permutation may lead to the permuted code with very poor SC performance. Indeed, let $n = 2^m$ be code length and $\mathbf{v}$ be a row of a constraint matrix. Consider a permutation $\pi \in \mathrm{GA}(m)$ with the matrix $\mathbf{P}$. Since $\mathbf{v} = \sum_{i=0}^{n-1} v_i\mathbf{e}_i$, by Remark 1 one has

$$\mathbf{v}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T = \sum_{i=0}^{n-1} v_i\mathbf{e}_i\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T = \sum_{i=0}^{n-1} v_i \sum_{i' \in \mathcal{G}_\pi(i)} \mathbf{e}_{i'}. \quad (6)$$

Therefore, $\mathbf{v}$ is transformed into the vector $\mathbf{v}'$ such that $v'_j = 1$ iff j belongs to symmetric difference of all $\mathcal{G}_\pi(i)$ with $v_i = 1$. Let i' be the "new" frozen symbol index, i.e. $i' = \mathrm{end}(\mathbf{v}')$. When $\mathbf{v}$ represents a static freezing constraint, $\mathbf{v} = \mathbf{e}_i$ for some $i \in \mathcal{F}$, hence $i' = \max \mathcal{G}_\pi(i)$ and by Remark 2 $\mathrm{wt}(i) = \mathrm{wt}(i')$. Recall that reliability of a virtual subchannel depends on the Hamming weight of its index. Thus, such transform of $\mathbf{v}$ does not entail freezing of some information symbol which is significantly more reliable than u_i. However, if $\mathbf{v}$ defines a DFC, there might be positions of ones in $\mathbf{v}$ of higher Hamming weight than the frozen index. Then it is possible that $\mathrm{wt}(i') > \mathrm{wt}(i)$ and besides ones in the BE of i' are concentrated and high bit positions. According to the partial order [20], [21], such index i' corresponds to much less noisy symbol that the i-th one. As the result, the reliable information bit $u_{i'}$ may turn to be frozen and some noisy bit (which is frozen in the original code) becomes information, increasing P_{SC} in terms of (1).

Example 3: Consider the $(16, 9)$ randomized polar subcode with $f_A = 2,\ f_B = 0$. The base code is defined by the frozen set $\{0, 1, 2, 4, 8\}$. The dynamic frozen symbols are u_{10} and u_{12} with possible constraints

$$u_{10} = u_3 + u_9,$$
$$u_{12} = u_5 + u_9 + u_{11}.$$

Let $\pi : (\mathbf{A}, \mathbf{b}) \in \mathrm{GA}(4)$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Since $\det \mathbf{A}_{0,1,3}^{1,2,3} \ne 0$, by Theorem 1 $2^1 + 2^2 + 2^3 = 14 \in \mathcal{G}_\pi(11)$. Meanwhile, $\mathcal{G}_\pi(12) = \{6, 8, 10, 12\}$ and therefore the DFC on u_{12} brings to the permuted code the frozen symbol u_{14} which is far more reliable than u_{12}. The frozen set of the permuted code is $\{0, 1, 2, 4, 8, 10, 14\}$ and u_{12} is unfrozen.

Consequently, we note that for any $j \in \mathcal{I}_m$ there exists $\pi \in \mathrm{GA}(m)$ such that in the BE of $j' = \max \mathcal{G}_\pi(j)$ the ones are set at maximal positions: $j'_{m-r-1} = j'_{m-r} = \dots = j'_{m-1} = 1$, where $r = \mathrm{wt}(j) = \mathrm{wt}(j')$. In fact, one can take the factor matrix with a non singular submatrix spanned by rows with indexes of ones in the BE of j and columns $m - r - 1, m - r, \dots, m - 1$ (as illustrated in Example 3). Hence, for a row $\mathbf{v}$ of a constraint matrix, there always exists a GA permutation which makes the maximal index of weight $\max_{j: v_j = 1} \mathrm{wt}(j)$ frozen in the permuted code. We also observe that an arbitrary affine permutation often transforms numerous DFCs into constraints on some reliable initially unfrozen symbols, resulting in poor SC decodability of the permuted code.

To overcome this issue, we propose to use permutations π from a subgroup of $\mathrm{GA}(m)$, which transform a DFC system in more limited way. Our main idea is to guarantee the order between elements of $\mathcal{G}_\pi(j)$ and $\mathcal{G}_\pi(i)$ where in some row $\mathbf{v}$ of a constraint matrix $i = \mathrm{end}(\mathbf{v})$ and $v_j = 1$, i.e. the value of u_i dynamically depends on u_j. More formally, we require

$$\forall j' \in \mathcal{G}_\pi(j)\ \exists i' \in \mathcal{G}_\pi(i) \text{ such that } j' < i'. \tag{7}$$

Proposition 2: If $i = \mathrm{end}(\mathbf{v})$ and (7) holds whenever $j < i$ and $v_j = 1$, then $\max \mathcal{G}_\pi(i) = \mathrm{end}(\mathbf{v}')$ where $\mathbf{v}' = \mathbf{v}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$ and $\mathbf{P}$ is the matrix of π.

Proof: Let $i^* = \max \mathcal{G}_\pi(i)$. If $i^* \in \mathcal{G}_\pi(j)$ for some j s.t. $j < i,\ v_j = 1$, then i^* is not maximal among $\mathcal{G}_\pi(i)$ by (7). Hence, $\mathbf{e}_{i^*}$ is present in the final sum in (6) exactly once, so $v'_{i^*} = 1$. And for any summand $\mathbf{e}_{j'},\ j' \in \mathcal{G}_\pi(j),\ j < i,\ v_j = 1$ there is $i' \in \mathcal{G}_\pi(i)$ such that $j' < i' \le i^*$. □

We notice that (7) can be ensured for any affine permutation π if any BE of a frozen index i covers all BEs of symbol indices which u_i depends on. More precisely, when for all rows $\mathbf{v}$ of the constraint matrix $\mathbf{V}$ one has

$$v_j = 1 \Rightarrow j \sqsubseteq i = \mathrm{end}(\mathbf{v}), \tag{8}$$

(7) takes place by Lemma 4. However, this is very strict condition. According to the partial order [20], [21], the symbol u_j is noisier than u_i and often j is also frozen, so that there is another row of $\mathbf{V}$ which ends at j. Thus, there is an invertible transformation of rows of $\mathbf{V}$, after which every row is $\mathbf{e}_f$ for some $f \in \mathcal{F}$, so that (8) entails all DFCs to be trivial.

C. Block-Diagonal Affine Permutations

In this section, we consider the group $\mathrm{BDA}(\mathbf{s}) \subseteq \mathrm{GA}(m)$ for some $\mathbf{s} = [s_1, s_2, \dots, s_t]$ such that $\sum_{i=1}^t s_i = m$ (see Section II-C). In what follows $\mathbf{s}$ denotes such list of block sizes. We show that for BDA permutations (7) can be guaranteed by a much flexible condition on the DFC system rather than (8). So, there exist non-trivial polar subcodes whose DFCs fulfill (7) for some BDA group. Besides, we show that BDA permutations preserve P_{SC} much better than arbitrary affine ones leading to permuted codes with pretty similar SC decodability as an original code.

In fact, if $\pi \in \mathrm{BDA}(\mathbf{s})$, the relation $\sqsubset$ in Lemma 4 can be replaced with a more specific partial order introduced below.

Definition 4: Let $\zeta_p = \sum_{i=1}^{p-1} s_i,\ 1 \le p \le t$ be beginning index of the p-th block. Then for $\alpha, \beta \in \mathcal{I}_m$

1) $\alpha^{(p)} = (\alpha_{\zeta_p}, \alpha_{\zeta_p+1}, \dots, \alpha_{\zeta_p+s_p-1})$ is the p-th block in the BE of α;
2) $\alpha <_\mathbf{s} \beta$ iff there exists $p:\ 1 \le p \le t$ such that $\alpha^{(p)} \sqsubset \beta^{(p)}$ and for all $q = p+1, \dots, t$ one has $\alpha^{(q)} = \beta^{(q)}$.

To get less formal definition of $<_\mathbf{s}$, consider $\mathbf{s} = [1, 1, \dots, 1]$. Then $<_\mathbf{s}$ is exactly the relation $<$. Otherwise $<_\mathbf{s}$ performs standard numeric comparison up to "merging" some digits into blocks and these blocks are compared using $\sqsubset$. Since $\sqsubset$ is defined partially, $<_\mathbf{s}$ is a strict partial order as well.

Corollary 1: Let $\alpha, \beta \in \mathcal{I}_m,\ \alpha <_\mathbf{s} \beta,\ \pi \in \mathrm{BDA}(\mathbf{s})$. Then for all $\alpha' \in \mathcal{G}_\pi(\alpha)$ there exists $\beta' \in \mathcal{G}_\pi(\beta)$ such that $\alpha' <_\mathbf{s} \beta'$.

Proof: Let $\pi : (\mathbf{A}, \mathbf{b})$. Given

$$\mathbf{A} = \begin{pmatrix} \mathbf{B}_1 & 0 & \dots & 0 \\ 0 & \mathbf{B}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{B}_t \end{pmatrix},$$

π performs independent blockwise affine transformations $\sigma_i : (\mathbf{B}_i, \mathbf{b}^{(i)})$ where $\mathbf{b}^{(i)}$ is the i-th block of $\mathbf{b}$. More precisely, the BE of α' has the form $(\gamma_{1,0}, \gamma_{1,1} \dots, \gamma_{1,s_1-1}, \dots, \gamma_{t,0}, \gamma_{t,1}, \dots, \gamma_{t,s_t-1})$ where $(\gamma_{i,0}, \gamma_{i,1}, \dots, \gamma_{i,s_i-1})$ is BE of some $\gamma_i \in \mathcal{G}_{\sigma_i}(\alpha^{(i)})$. By definition of $<_s$ there is p such that $\alpha^{(p)} \sqsubset \beta^{(p)}$ and $\alpha^{(q)} = \beta^{(q)}\ \forall q = p+1, \dots, t$, so $\mathcal{G}_{\sigma_q}(\alpha^{(q)}) = \mathcal{G}_{\sigma_q}(\beta^{(q)})\ \forall q = p+1, \dots, t$. By Lemma 4, there is $\delta_p \in \mathcal{G}_{\sigma_p}(\beta^{(p)})\gamma_p \sqsubset \delta_p$. Finally, take β' with BE $\langle \delta_1, \dots, \delta_p, \gamma_{p+1}, \dots, \gamma_t \rangle$, where $\langle \cdot, \dots, \cdot \rangle$ is concatenation of BEs and for $q = 1, \dots, p-1 \delta_q$ is an arbitrary element of $\mathcal{G}_{\sigma_q}(\beta^{(q)})$. This gives rise to $\alpha' <_s \beta'$ by definition while obviously $\beta' \in \mathcal{G}_\pi(\beta)$. □

Thus, if $j <_\mathbf{s} i$ and $\pi \in \mathrm{BDA}(\mathbf{s})$, (7) is satisfied (because $j' <_\mathbf{s} i'$ implies $j' < i'$). Now the difference in reliability between the symbols u_i and $u_{i'}, i' \in \mathcal{G}_\pi(i)$ is to be investigated.

Definition 5: Let $\alpha, \beta \in \mathcal{I}_m$. Then $\alpha \sim_\mathbf{s} \beta$ iff for all p such that $1 \le p \le t$ one has $\mathrm{wt}(\alpha^{(p)}) = \mathrm{wt}(\beta^{(p)})$.

Corollary 2: Let $\alpha \in \mathcal{I}_m,\ \pi \in \mathrm{BDA}(\mathbf{s})$. Then $\max(\mathcal{G}_\pi(\alpha)) \sim_\mathbf{s} \alpha$.

Proof: Consider $\alpha' \in \mathcal{G}_\pi(\alpha)$ and the block decomposition $(\sigma_1, \sigma_2, \dots, \sigma_t),\ \sigma_i \in \mathrm{GA}(s_i)$ for π and $(\gamma_1, \gamma_2, \dots, \gamma_t),\ \gamma_i \in \mathcal{G}_{\sigma_i}(\alpha^{(i)})$ for the BE of α' from the previous proof. For some $p:\ 1 \le p \le t$ let $\gamma_p^* = \max \mathcal{G}_{\sigma_p}(\alpha^{(p)})$ and $\beta \in \mathcal{G}_\pi(\alpha)$ is obtained from α' by replacing its p-th block of the BE (i.e. γ_p) with γ_p^*. If $\gamma_p \ne \gamma_p^*$, then $\gamma_p < \gamma_p^*$ and $\alpha' < \beta$. So, if $\alpha' = \max \mathcal{G}_\pi(\alpha)$, then $\gamma_p = \gamma_p^*$ for every p. By Remark 2 (applied to $\alpha^{(p)}$ and σ_p) one has $\mathrm{wt}(\gamma_p^*) = \mathrm{wt}(\alpha^{(p)})$ what terminates the proof. □

Let us summarize the current results. Consider a permutation matrix $\mathbf{P}$ corresponding to $\pi \in \mathrm{BDA}(\mathbf{s})$, a polar subcode $\mathcal{C}$ with constraint matrix $\mathbf{V}$ and the permuted code $\mathcal{C}'$. The rows of constraint matrix of $\mathcal{C}'$ can be obtained by applying (6) to every row $\mathbf{v}$ of $\mathbf{V}$. Next, consider the DFC which is defined by $\mathbf{v}$: $u_i = \sum_{j \in J} u_j$, where $i = \mathrm{end}(\mathbf{v})$, $J = \{j < i \mid v_j = 1\}$. If for all $j \in Jj <_\mathbf{s} i$, then $\mathrm{end}(\mathbf{v}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T) \in \mathcal{G}_\pi(i)$ Proposition 2 and Corollary 1.

Example 4: Consider a code of length $n = 32$ and the row $\mathbf{v}$ corresponding to the static frozen symbol u_{20}: $v_{20} = 1$ and $v_i = 0\ \forall i \ne 20$. For $\mathbf{s} = [2, 3]$ all indices from $I = \{0, 1, \dots, 7, 16, 17, 18, 19\}$ are in the relation $<_\mathbf{s}$ with 20. And by Corollary 1 setting $v_i = 1$ for any $i \in I$ preserves $\mathrm{end}(\mathbf{v}\mathbf{G}_5^T\mathbf{P}\mathbf{G}_5^T) = \max \mathcal{G}_\pi(20)$ when $\mathbf{P}$ is matrix of arbitrary $\pi \in \mathrm{BDA}(2, 3)$. For the case of rate-1/2 code and BI-AWGN channel with $E_b/N_0 = 3$ dB, $\hat{P}_{20} \approx 6.5 \cdot 10^{-2}$. By Corollary 2 the "new" frozen symbol $\max \mathcal{G}_\pi(20)$ belongs to $\{12, 20, 24\}$. Note that the bits u_{12} and u_{24} are also unreliable since $\hat{P}_{12} \approx 0.127,\ \hat{P}_{24} \approx 3.89 \cdot 10^{-2}$ and the error probability in the original channel is approximately $4.58 \cdot 10^{-2}$.

As illustrated in the example above, we often observe that for $i \in \mathcal{F},\ i' = \max \mathcal{G}_\pi(i)$ one has $\hat{P}_{i'} \approx \hat{P}_i$. The reason is that the reliability of a subchannel depends on the Hamming weight of its BE, especially in the case of high signal-to-noise-ratio (SNR) [24]. On the other hand, for indices of equal Hamming weight, the reliability strongly depends on how big the positions of ones in the BE are. This implies that if the i-th subchannel is unreliable, the i'-th one is also unreliable as long as the block sizes (items of $\mathbf{s}$) are not greater than some fixed not very large integer S.

IV. Polar Subcodes for Permutation Decoding

In this section two families of polar subcodes for permutation-based decoding under block-diagonal affine permutations are presented. As before, $\mathbf{s}$ denotes a list of block sizes.

A. Randomized Polar Subcodes

We propose to design DFC system of a RP subcode satisfying the condition of Proposition 2 when $\pi \in \mathrm{BDA}(\mathbf{s})$. For this, we select DFC system that ensures the relation $<_{\mathbf{s}}$ between a position of non-zero element in constraint matrix and the corresponding frozen index. On the other hand, one should minimize SC error probability for images of a base code under BDA permutations, because most of freezing constraints of a RP subcode (namely, all constraints except f_A+f_B equations) are inherited from base code.

To begin with, we define $\mathrm{P}(s_1, s_2, \ldots, s_t) \triangleq \mathrm{P}(\mathbf{s})$ as a group of affine permutations with a block-permutation factor matrix and zero shift. Its action on $\mathcal{I}_m$ leads to the set $\mathcal{I}_m/\mathrm{P}(\mathbf{s})$ consisting of equivalence classes under relation "$\sim$": $i \sim j$ iff there exists $\pi \in \mathrm{P}(\mathbf{s})$ such that $i = \pi(j)$. It can be easily checked that "$\sim$" and "$\sim_{\mathbf{s}}$" are exactly the same relation, so $i \sim_{\mathbf{s}} j$ iff i and j belong to the same class. For instance, $\mathcal{I}_4/\mathrm{P}(2,2) = \{\{0\},\{1,2\},\{3\},\{4,8\},\{5,6,9,10\},\{7,11\},\{12\},\{13,14\},\{15\}\}$. It is suggested in [11] to build frozen set of a polar code as union of equivalence classes. Then automorphism group includes $\mathrm{P}(\mathbf{s})$. For our purpose it is not necessary to build a permutation-invariant base code: it is sufficient to make frozen sets of original code and permuted codes identical. A polar code whose frozen set is a union of classes from $\mathcal{I}_m/\mathrm{P}(\mathbf{s})$ may not be a decreasing monomial code [20], but the frozen set is preserved under permutations from $\mathrm{BDA}(\mathbf{s})$.

Theorem 2: Consider a polar code $\mathcal{C}$ whose frozen set $\mathcal{F}$ satisfies

$\forall i \in \mathcal{F}$ and $j \in \mathcal{I}_m$ such that $j \sim_{\mathbf{s}} i$ one has $j \in \mathcal{F}$, $\pi \in \mathrm{BDA}(\mathbf{s})$, $\mathcal{C}' = \{\pi\mathbf{c} \mid \mathbf{c} \in \mathcal{C}\}$. Then $\mathcal{F}$ is frozen set of $\mathcal{C}'$.

Proof: of this statement stems from the following lemma.

Lemma 5: Let $\mathbf{V} \in \mathbb{F}_2^{l\times n}$, $n = 2^m$, $\varepsilon_i = \mathrm{end}(\mathbf{V}_i)$, $0 \le i < n$. We say that $\mathbf{V}$ is normal if

(i) All ε_i are distinct.
(ii) For all $i \in \{0, 1, \ldots, l-1\}$ and $j < \varepsilon_i$ the statement $v_{i,j} = 1$ implies $j \sqsubset \varepsilon_i$.

Let $\mathbf{P}$ be matrix of a permutation $\pi \in \mathcal{S}_n$, $\mathbf{W} = \mathbf{V}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$, $\varepsilon_i' = \mathrm{end}(\mathbf{W}_i)$. Then, supposing that $\mathbf{V}$ is normal,

1) if $\pi \in \mathrm{P}(m)$, $\mathbf{W}$ is normal and $\varepsilon_i' = \pi(\varepsilon_i)$;
2) if $\pi \in \mathrm{LTA}(m)$, $\mathbf{W}$ is normal and $\varepsilon_i' = \varepsilon_i$.

Proof: In the case 1 for every $i \in \mathcal{I}_m$ $\mathcal{G}_\pi(i)$ contains single element $\pi(i)$. Let $j' \in \mathcal{I}_m$: $j' \ne \pi(\varepsilon_i)$ and $w_{i,j'} = 1$. By (6) there is $j \in \mathcal{I}_m$ such that $v_{i,j} = 1$ and $j' \in \mathcal{G}_\pi(j)$, i.e. $j' = \pi(j)$. Since $j \ne \varepsilon_i$ (otherwise $j' = \pi(\varepsilon_i)$), $j \sqsubset \varepsilon_i$. Provided π permutes digits in a binary expansion, $\pi(\varepsilon_i)$ has ones at all the positions of ones in j', so $j' \sqsubset \pi(\epsilon_i)$ and $\varepsilon_i' = \pi(\varepsilon_i)$. Thus, (ii) is proven. To prove (i) recall that π acts bijectively and $(\varepsilon_i)_i$ are distinct, so $(\varepsilon_i')_i$ are distinct as well.

From [20, Proof of Theorem 2] one can conclude that in the case 2 $\forall i \in \mathcal{I}_m$ $i \in \mathcal{G}_\pi(i)$ and all elements of $\mathcal{G}_\pi(i)$ are in the relation $\sqsubseteq$ with i. One has that if $j < \varepsilon_i$, $j' \in \mathcal{G}_\pi(j)$ and $v_{i,j} = 1$, then $j' \le j < \varepsilon_i$ while $\varepsilon_i \in \mathcal{G}_\pi(\varepsilon_i)$. Hence, $\varepsilon_i' = \varepsilon_i$ (what also proves (i)) and besides $j' \sqsubseteq j$ and $j \sqsubset \varepsilon_i$ (by condition (ii)), so $j' \sqsubset \varepsilon_i$. □

Proof: [**Proof of Theorem 2**] It is obvious that the constraint matrix of $\mathcal{C}$ is normal in terms of Lemma 5. In fact, every its row has single non-zero element at the position of the corresponding frozen symbol. As shown in [12], π can be presented as a composition of permutations from $\mathrm{P}(\mathbf{s}) \subseteq \mathrm{P}(m)$ and $\mathrm{LTA}(m)$. Their successive application keeps constraint matrix of the permuted code normal by Lemma 5. We need to prove that in both cases set of row ends is not being changed. If a permutation from $\mathrm{P}(\mathbf{s})$ is applied, it maps every row end into another row end, since row ends are frozen indices and $\mathcal{F}$ is invariant with respect to $\mathrm{P}(\mathbf{s})$. In case of $\mathrm{LTA}(m)$, end of every row is kept unchanged. □

A method for construction of polar codes with digit permutation invariant frozen set that uses knapsack problem solution was suggested in [11]. We present a modification of this method of constructing RP subcodes for permutation decoding. Given code length $n = 2^m$, dimension k, block structure $\mathbf{s}$ and design parameters f_A, f_B (described in Section II-B), codes obtained by the following algorithm can be efficiently decoded if permutations from $\mathrm{BDA}(\mathbf{s})$ are used.

1) Build a base $(n, k+f_A)$ polar code solving the following knapsack problem:
   - Items are equivalence classes $C \in \mathcal{I}_m/\mathrm{P}(\mathbf{s})$ of indices from $\mathcal{I}_m$ under the relation "$\sim_{\mathbf{s}}$".
   - Weight of an item C is $|C|$.
   - Cost of an item C is $\sum_{i\in C}\log(1-\hat{P}_i)$, total cost is being minimized.
   - Total weight is not greater than $n-k-f_A$.

   So, the frozen set of the base code is $\mathcal{F} = \bigcup_j C_j$ where the classes $\{C_j\}_j$ form the solution.
2) Apply random DFCs using parameters f_A and f_B as described in Section II-B or in [16, Section III-D]
   - Let $r = \min_{i\in\mathcal{A}} \mathrm{wt}(i)$ and $I = \{i_1, i_2, \ldots, i_{f_A}\}$ be the set of maximal elements in $\mathcal{A}$ of Hamming weight r. If there are less than f_A elements in $\mathcal{A}$ of such weight, add to I maximal elements of weight $r+1, r+2$, and so on, until $|I| = f_A$. Then for every $l = 1, \ldots, f_A$ set symbol u_{i_l} dynamic frozen: $u_{i_l} = \sum_{p=0}^{i_l-1} a_{l,p}u_p$ where $(a_{l,p})_p$ are random independent equiprobable binary values.
   - Let $J = \{j_1, j_2, \ldots, j_{f_B}\} \subseteq \mathcal{F}$ be the set of indices of the most reliable frozen symbols of the base code. Then, as previously, for $l = 1, \ldots, f_B$ set u_{j_l}

dynamic frozen: $u_{j_l} = \sum_{p=0}^{j_l-1} b_{l,p}u_p$ where $(b_{l,p})_p$ are random independent equiprobable binary values.

3) For every DFC exclude from right hand side of its equation $u_i = \sum_j u_j$ all u_j such that $j \not<_{\mathbf{s}} i$. More formally, for every frozen symbol u_i one needs to set $v_{t_i,j} = 0$ for all $j < i$ such that $j \not<_{\mathbf{s}} i$ where $\mathbf{V}$ is the constraint matrix of the code obtained after the previous step and $i = \operatorname{end}(\mathbf{V}_{t_i})$.

The only difference between the method suggested in step 1 for constructing an invariant polar code and the method from [11] is the definition of an item cost. The suggested costs are needed to minimize the value of P_{SC} since

$$\begin{aligned}\phi(\mathcal{F}) &\triangleq \sum_{i\in\mathcal{F}} \log(1-\hat{P}_i)\\ &= \log \prod_{i\in\mathcal{F}}(1-\hat{P}_i)\\ &= \log \prod_{i\in\mathcal{I}_m}(1-\hat{P}_i) - \log\prod_{i\in\mathcal{A}}(1-\hat{P}_i).\end{aligned}$$

Here the first term is a constant while the second one is $h(p) = -\log(1-p)$ where p is the approximation of P_{SC} from (1). Provided h is a monotonically growing function, minimization of $\phi(\mathcal{F})$ performed by knapsack problem solution leads to optimization of P_{SC} in terms of (1).

The purpose of step 3 is to fulfill the condition of Corollary 1. The example below illustrates application of this step.

Example 5: Consider construction of the $(16, 9)$ code for $f_A = 2,\ f_B = 0$ and $\mathbf{s} = [1, 3]$. Its frozen set and possible DFCs are given in Example 3. To make the constraint system compliant with permutation decoding, one should exclude the summand u_{11} from the equation for u_{12} since $11 \not<_{\mathbf{s}} 12$ while other terms in the DFCs satisfy the relation $<_{\mathbf{s}}$.

Let us examine frozen sets of permuted codes when the original code is constructed according to the algorithm above. We recommend to set $f_B = 0$: in this case all static frozen symbols of the original code are frozen in the permuted codes. Indeed, in this case the constraint matrix has the form

$$\mathbf{V} = \begin{pmatrix}\overline{\mathbf{V}}\\ \tilde{\mathbf{V}}\end{pmatrix},$$

where $\overline{\mathbf{V}}$ is the constraint matrix of the base code $\overline{\mathcal{C}}$ and rows of $\tilde{\mathbf{V}}$ correspond to type-A DFCs. Since the frozen set $\overline{\mathcal{F}}$ of $\overline{\mathcal{C}}$ is invariant with respect to $\mathrm{P}(\mathbf{s})$, $\overline{\mathcal{F}}$ is also the frozen set of the code $\overline{\mathcal{C}}' = \{\pi\mathbf{c} \mid \mathbf{c} \in \overline{\mathcal{C}}\},\ \pi \in \operatorname{BDA}(\mathbf{s})$ by Theorem 2. Provided the constructed code permuted with π is a subcode of $\overline{\mathcal{C}}'$, all indices from $\overline{\mathcal{F}}$ are kept frozen.

However, the set of type-A dynamic frozen symbol (DFS-A) indices may not be invariant with respect to $\mathrm{P}(\mathbf{s})$. But since they are maximal among all elements of $\mathcal{I}_m$ of some weight, ones in their binary expansions are set at high bit positions. So, for every DFS-A u_i almost all unfrozen symbols $u_j,\ j \sim_{\mathbf{s}} i$ (by Corollary 2 only such j may be maximal among $\mathcal{G}_\pi(i),\ \pi \in \operatorname{BDA}(\mathbf{s})$) are less reliable than u_i. Moreover, it is possible that many rows of $\mathbf{W} = \mathbf{V}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$ (where $\mathbf{P}$ is matrix of $\pi \in \operatorname{BDA}(\mathbf{s})$) end at the same position (let it be i'). After Gaussian elimination there may be a frozen symbol with an index from $\mathcal{G}_\pi(j)$ where u_j is reliable unfrozen symbol of the original code, so P_{SC} is increasing. Nevertheless, in the row echelon form of $\mathbf{W}$ there is a row $\mathbf{w}$ such that $\operatorname{end}(\mathbf{w}) = i'$, so at least this row corresponds to an unreliable (since $i' \in \mathcal{G}_\pi(i)$ where $i \in \mathcal{F}$) frozen symbol. And of course, there are only $f_A + f_B$ DFCs in the original code that may lead to such "problem" rows in $\mathbf{W}$.

B. Polar Subcodes of e-BCH Codes

Another way to exploit knapsack problem technique is to build polar subcodes of e-BCH codes for permutation decoding. As for the standard e-BCH subcodes [4], the proposed construction inherits design minimum distance of a parent e-BCH code since the constraint system includes the parent DFCs. In addition, one needs to have the noisy message bits frozen in order to make the code appropriate for SC decoding. The presented construction is similar to that from Section IV-A, where good SC decodability of permuted codes is assured by digit permutation invariance of the static frozen index set.

The key idea of the proposed e-BCH construction is to make frozen set invariant with respect to $\sim_{\mathbf{s}}$ up to some frozen symbols of a parent code. For this, we suggest to select the set of indices of static frozen symbols, added to constraint system of a parent e-BCH code, as a union of classes from $\mathcal{I}_m/\mathrm{P}(\mathbf{s})$. Some elements of classes that are included may be already frozen in a parent code. Then such additional static freezing constraints are not performed. More precisely, one needs to solve the following optimization problem. Given a parent e-BCH code of length $n = 2^m$ with frozen set $\overline{\mathcal{F}}$, virtual subchannel error probabilities $(\hat{P}_i)_{i\in\mathcal{I}_m}$, desired dimension k and a list of block sizes $\mathbf{s}$, it is needed to find $\mathcal{F} \supseteq \overline{\mathcal{F}}$ such that:

1) $|\mathcal{F}| \le n - k$.
2) For all $i \in \mathcal{F} \setminus \overline{\mathcal{F}}$ and $j \sim_{\mathbf{s}} i$ one has $j \in \mathcal{F}$.
3) $\sum_{i\in\mathcal{F}} \log(1-\hat{P}_i)$ is being minimized.

For a design distance d we present the following method of constructing polar subcodes of e-BCH codes:

1) Build an (n, K, d) e-BCH code $\overline{\mathcal{C}}$ with constraint matrix $\overline{\mathbf{V}}$. For that purpose consider the parity check matrix of $\overline{\mathcal{C}}$ over the field $\mathbb{F}_{2^m}$:
$$\mathbf{H} = \begin{pmatrix} x_0^0 & x_1^0 & \dots & x_{n-1}^0\\ x_0^1 & x_1^1 & \dots & x_{n-1}^1\\ \vdots & \vdots & \ddots & \vdots\\ x_0^{d-2} & x_1^{d-2} & \dots & x_{n-1}^{d-2}\end{pmatrix}, \qquad (9)$$
where $x_i = \sum_{j=0}^{m-1} i_j\alpha^j$ and α is a primitive element of the field. One needs to expand elements of $\mathbf{H}\mathbf{G}_m^T$ to binary columns according to their representation in the standard basis of $\mathbb{F}_{2^m}$ and apply Gaussian elimination (removing linearly dependent rows) what leads to $\overline{\mathbf{V}}$. Indeed, $\overline{\mathcal{C}} = \{\mathbf{c} \in \mathbb{F}_2^n \mid \mathbf{c}\mathbf{H}^T = 0\}$, so the vector $\mathbf{u}\mathbf{G}_m,\ \mathbf{u} \in \mathbb{F}_2^n$ is a codeword iff $\mathbf{u}\overline{\mathbf{V}}^T = 0$. Let $\overline{\mathcal{F}}$ be the frozen set which can be easily obtained from $\overline{\mathbf{V}}$.
2) Solve the knapsack problem:
   - Items are sets of the form $C \setminus \overline{\mathcal{F}},\ C \in \mathcal{I}_m/\mathrm{P}(\mathbf{s})$.
   - Weight of an item M is $|M|$.

- Cost of an item M is $\sum_{i\in M}\log(1-\hat{P}_i)$, total cost is being minimized.
- Total weight is not greater than $K-k$.

Let $M_1,\ldots,M_l$ form the solution, then one needs to set static frozen all elements of $\bigcup_j M_j$. More formally, add to $\overline{\mathbf{V}}$ all the rows of the form $(0,\ldots,0,\underset{(i)}{1},0,\ldots,0)\in\mathbb{F}_2^n$ where $i\in M_j$ for some $j=1,\ldots,l$. The derived matrix is the constraint matrix of the resulting code.

As stated in [4], indices in DFCs of a parent e-BCH code have low Hamming weight. Their weights are preserved "blockwise" after a block-diagonal affine transformation according to Corollary 2. It means that for every row $\mathbf{v}$ of the constraint matrix $\mathbf{V}$ corresponding to some DFC, $\pi\in\mathrm{BDA}(\mathbf{s})$ with matrix $\mathbf{P}$ and $\mathbf{v}'=\mathbf{v}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$ one has that $v_j'=1$ implies j either to be in the relation $\sim_{\mathbf{s}}$ with some i such that $v_i=1$ or to be covered by some index which is in this relation. Hence, after Gaussian elimination in $\mathbf{V}\mathbf{G}_m^T\mathbf{P}\mathbf{G}_m^T$ all row ends either correspond to static frozen symbols of the original code or have low Hamming weight.

For both suggested design algorithms knapsack problem solution may return a frozen set with smaller cardinality than desired. But it is explained in [11] how to deal with such problem: one should add to frozen set of obtained k'-dimensional ($k'>k$) code $k'-k$ indices of less reliable unfrozen symbols regardless their equivalence classes belonging. In other words, the considered technique finds the best invariant code $\mathcal{C}$ among all the codes with dimension $\geq k$ and if it is needed to construct an exactly k-dimensional code, the subcode of $\mathcal{C}$ can be returned (obviously, decoding error probability for this subcode is less than for $\mathcal{C}$). Besides, in case of small block sizes equivalence classes are pretty small as well, so frozen set of exactly desired size often can be obtained. More precisely, the cardinality of an equivalence class containing some $i\in\mathcal{I}_m$ is

$$\binom{s_1}{r_1}\binom{s_2}{r_2}\cdots\binom{s_t}{r_t},\tag{10}$$

where $r_p=\mathrm{wt}(i^{(p)})$. If $r_p=0$ or $r_p=s_p$ (which is always true if $s_p=1$), the p-th multiplier in (10) can be omitted. For example, if $n=1024$ and in s two elements are 3 while the others are ones, there are 64 classes of size 9, 128 classes of size 3 and 64 classes of size 1, although $\mathrm{BDA}(\mathbf{s})$ often contains large number of pairwise non-equivalent permutations. If $\mathbf{s}$ has only one element $s_p=3$ while the others are ones, there are 512 classes: 256 of size 3 and 256 of size 1.

V. Numeric Results

In this section we present some results for codes constructed for BI-AWGN channel with $E_b/N_0=2$ dB.

In Table I we compare permuted versions of different RP subcodes when block-diagonal affine permutation is performed. In the experiment for $\mathbf{s}=[3,1,1,1,1,3]$ 10 permutations from $\mathrm{BDA}(\mathbf{s})$ were generated including the trivial one. Three code design algorithms (namely algorithm from [16] and two versions of the suggested algorithm: with and without step 3) were considered and 50 codes $(1024,512)$ with $f_A=16, f_B=48$ for each algorithm were generated. After that, for every code 10 its permuted versions were obtained. The permuted codes were used to calculate for every design algorithm average value of (1) for $E_b/N_0=2$ dB ($\overline{P}_{SC}$), average maximum weight among frozen set ($\overline{w}_{\max}$) and $\overline{N}_{w>5}$, average number of frozen indices of weight >5 (for that code parameters frozen symbols of weight >5 often are most reliable among $\mathcal{F}$). Although the second design algorithm optimizes values of P_{SC} of permuted base codes, it has no gain in contrast with default RP subcodes. This is because there are a lot of frozen symbols of high Hamming weight arising from such summands u_j in DFCs that $j\not\prec_{\mathbf{s}} i$ where u_i is dynamic frozen symbol. It can be seen that the third algorithm provides smaller values of $\overline{P}_{SC}$, $\overline{w}_{\max}$ and $\overline{N}_{w>5}$ compared to the second one. This shows that eliminating such "dangerous" summands from random DFCs significantly reduces both maximal and average weight of frozen symbols what guarantees better SC (or SCL) decodability of permuted codes. The last row shows that setting $f_B=0$ for the proposed construction provides permuted codes with even better configurations of frozen symbols.

TABLE I
Frozen Symbol Weights of Permuted $(1024,512)$ Randomized Polar Subcodes

Code design		$\overline{P}_{SC}$	$\overline{w}_{\max}$	$\overline{N}_{w>5}$
RP from [16]		0.923	8.7	45.218
Suggested RP	without step 3	0.945	8.4	45.854
	with step 3, $f_B\neq 0$	0.794	6.582	21.444
	with step 3, $f_B=0$	0.713	6.438	16.728

TABLE II
Error Coefficient of $(1024,512)$ Randomized Polar Subcodes

f_A	RP from [16], $f_B=64-f_A$			Suggested RP, $f_B=0$		
	A'_{16}	A_{16}	P_{SC}	A'_{16}	A_{16}	P_{SC}
0	41152	33023	0.165	27840	27840	0.561
6	53439	609	0.204	27840	367	0.65
9	54644	70	0.226	27840	102	0.669
10	54644	23	0.233	47680	85	0.669
11	54644	11	0.241	42176	143	0.698
12	58560	5	0.248	42176	121	0.698
13	66752	0	0.253	76352	114	0.698
14	66752	0	0.261	42176	63	0.734
15	66752	0	0.269	42176	39	0.734
16	66752	0	0.276	76352	34	0.734
17	66752	0	0.28	27840	3	0.754
18	74944	0	0.285	27840	0	0.746

In Table II we consider distance properties of RP subcodes with $n=1024$ and $k=512$. Comparing the RP construction from [16] and the proposed one for $\mathbf{s}=[3,1,1,1,1,3]$, we report approximated values of P_{SC} at $E_b/N_0=2$ dB and the error coefficients (i.e. numbers of codewords with Hamming weight ≤ 16) of the base codes (A'_{16}) and the resulting RP subcodes (A_{16}). For each code we obtained A_{16} and A'_{16} using the algorithm from [25]. Although in the suggested design we use not all possible random DFCs (i.e. u_i may dynamically depend on u_j only if $j<_{\mathbf{s}} i$), the type-A DFCs still eliminate majority of low-weight codewords. We also notice that A'_{16} for the proposed codes may decrease when f_A grows. The reason is that we additionally require the frozen set of a base code to be $\mathrm{P}(\mathbf{s})$-invariant. Hence, for different values of $K=k+f_A$,

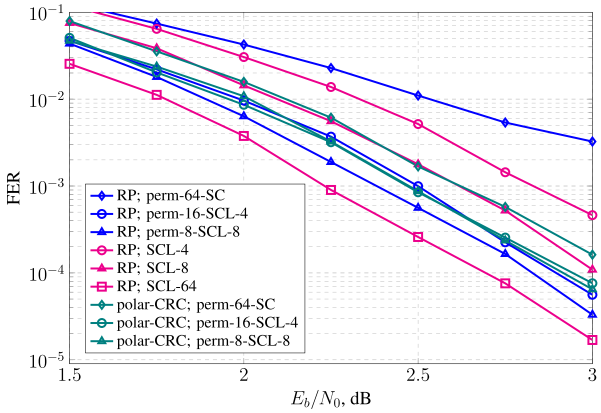

Fig. 3. Performance of $(256, 128)$ codes.

in general case the K-dimensional base code in not a subcode of the $(K+1)$-dimensional one. As a consequence, a RP subcode with lower value of f_A may provide better error coefficient (e.g. for $f_A = 10$ the value of A_{16} is lower than for $f_A = 11, 12, 13$). And because of more constrained choice of a base code, we observe much worse values of P_{SC} than for classical RP subcodes. Nevertheless, simulation results show that ensemble of parallel decoders of permuted codes demonstrate much better performance than a standard polar subcode with a decoder having similar complexity as each of component decoders.

Fig. 3 presents simulation results for $(256, 128)$ RP subcodes obtained using the proposed method, codes from [16] under SCL decoder and CRC-aided polar code (polar-CRC). For decoding of the proposed codes permutations were generated according to the heuristic from [13]. For every equivalence class of affine permutations its representative from BDA group was obtained by taking factor matrix as product of a block-permutation matrix and an upper-triangular matrix, while shift was selected randomly. In the experiment for every size of permutation list the special block sizes were used. Namely, for $M = 8$ they are $[1, 1, 1, 1, 1, 3]$, for $M = 16$: $[1, 2, 1, 1, 3]$ and $[3, 1, 1, 3]$ for $M = 64$. For moderate number of permutations ($8 \le M \le 64$) we suggest to take $s_p \le S = 3$ for all $p = 1, \dots, t$. Furthermore, simulation results show that the best way (in terms of FER) to select $\mathbf{s}$ is to make its elements as small as possible keeping desired number of equivalence classes (i.e. at least M) present in $\mathrm{BDA}(\mathbf{s})$. So, permutations from distinct random equivalence classes were picked (the trivial permutation is always present). Moreover, it was set $f_A = 8$, while $f_B = 32$ for RP subcodes from [16] and $f_B = 0$ for the proposed ones. To get the considered polar code with CRC, we take the decreasing monomial code $(256, 128)$ generated by minimal information set $\{31, 57\}$ [12]. Its automorphism group includes block lower triangular affine group with $\mathbf{s} = [3, 5]$ and in simulations permutations from $\mathrm{BDA}(3, 5)$ were generated in the same way as in [13]. This code is concatenated with 6-bit CRC, so permutation decoder based on SCL (SC is an instance of SCL with $L = 1$) merges lists of codewords obtained by component decoders, applies corresponding inverse permutations and returns the most likely candidate satisfying outer CRC.

TABLE III
COMPLEXITY OF DECODERS OF (256,128) CODES

Decoder	Average number of operations
perm-64-SC	$1.55 \cdot 10^5$
perm-16-SCL-4	$1.64 \cdot 10^5$
perm-8-SCL-8	$1.89 \cdot 10^5$
SCL-64	$2.76 \cdot 10^5$

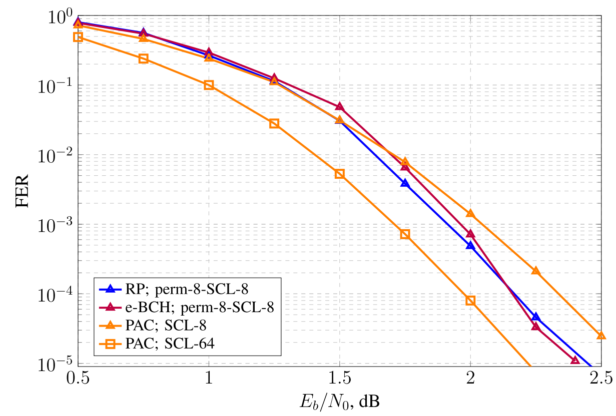

Fig. 4. Performance of $(1024, 512)$ PAC codes and proposed polar subcodes.

For RP subcodes permutation-based SCL decoding with small component decoder list size L expectedly outperforms standard SCL-L decoding. In comparison with a SCL-$(L \cdot M)$ decoder, the performance loss is ≈ 0.25 dB and gets lower at high-SNR region. Note that if SC decoder is used in the permutation algorithm (and value of $L \cdot M$ is kept constant), there is a significant performance degradation. The reason is that according to (1) SC error probability depends only on the frozen set, so gain of DFCs is not exploited by the component decoders. Furthermore, the proposed RP subcodes outperform the polar code with CRC. Note that effective dimension of the latter code is 122, to accommodate 6 bits of CRC.

Table III shows the complexity of the considered decoding algorithms. It can be seen that the complexity of the proposed decoding algorithm approaches that of the SCL decoder as L increases. Although one cannot use $L = 1$ in the proposed approach, one can still use it with small L and sufficiently many permutations M to obtain performance close to that of SCL-$(L \cdot M)$, as shown in Fig. 3. This allows one to avoid highly cumbersome implementation of SCL decoder with large list size.

Another construction of polar codes with dynamic frozen symbols is polarization-adjusted convolutional (PAC) codes [26]. In Fig. 4 we compare performance of the proposed $(1024, 512)$ RP subcode constructed for $f_A = 12,\ f_B = 0$ and $\mathbf{s} = [1, 1, 1, 1, 1, 1, 1, 3]$, the proposed $(1024, 512, 24)$ e-BCH subcode with $\mathbf{s} = [3, 1, 1, 1, 1, 3]$ and PAC codes from [27] that also have improved weight distribution. As in the previous case, we observe that the presented permutation decoding approach entails performance loss up to 0.25 dB that decreases with growing of E_b/N_0.

In Fig. 5 there is the performance comparison of $(2048, 1024)$ codes. Here for the proposed RP subcodes we use $f_A = 16,\ f_B = 0$ and $f_A = 16,\ f_B = 48$

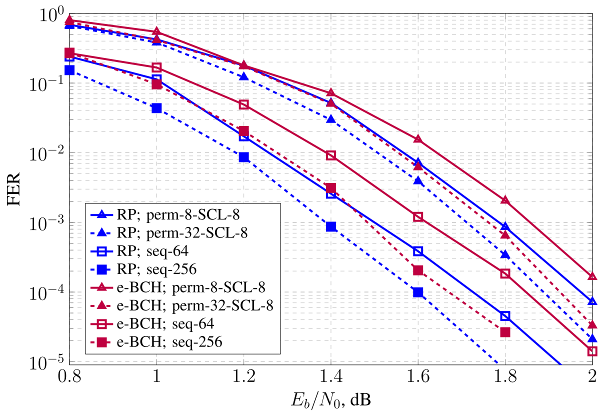

Fig. 5. Performance of $(2048, 1024)$ codes.

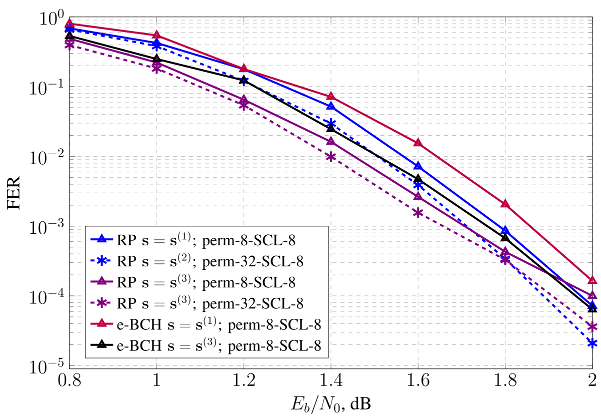

Fig. 6. Performance of proposed $(2048, 1024)$ polar subcodes with different block profiles.

for codes from [16], while e-BCH subcodes were constructed for the design distance $d = 40$. The block sizes are $\mathbf{s}^{(1)} = [1,1,1,1,1,1,1,1,3]$ for $M = 8$ and $\mathbf{s}^{(2)} = [1,2,1,1,1,1,1,3]$ for $M = 32$. As for short length, at high values of E_b/N_0 the loss of a permutation decoder is about 0.2-0.25 dB. Moreover, performance of the proposed decoder may be improved by increasing size of the permutation list keeping the parameter L for the component SCL decoders constant.

Finally, we note that altering the block structure $\mathbf{s}$ in code design and permutation selection may noticeably affect FER of the introduced decoding approach. In fact, in Fig. 6 we compare performance of codes considered in Fig. 5 and the codes constructed for $\mathbf{s}^{(3)} = [1,1,1,1,3,1,1,2]$ under permutation-based SCL-8 decoding. At low and moderate SNR region the RP subcodes with $\mathbf{s}^{(3)}$ exhibit the performance gain up to 0.2 dB. However, for $E_b/N_0 \approx 2$ dB they start loosing to the codes with $\mathbf{s}^{(1)}$ and $\mathbf{s}^{(2)}$. On the other hand, the e-BCH subcode with $\mathbf{s}^{(3)}$ outperforms the previously considered code at all values of E_b/N_0.

VI. Conclusion

In this paper an extension of the permutation decoding algorithm to the case of polar subcodes was presented. Since the affine automorphism group of a polar subcode is a difficult and currently unexplored object, the proposed decoders use permutations outside of it. Some properties of affine permutations were established. Based on these properties, the modified constructions of polar subcodes were introduced. However, these properties may be used in some other applications.

Simulations show that proper selection of parameters allows to attain error-rate performance near SCL or sequential decoder with a large list size. For this, it is enough to use in component decoders SCL algorithm with $L = 8$. The proposed approach may be effective in practice when sufficient FER cannot be achieved by standard SCL decoding with medium L. But, the results can be improved using more advanced techniques for permutation sampling, for example from [15] or [28].

Appendix

A. Proof of Lemma 1

Definition 6: For $f \in \mathcal{B}_m$ let $\deg f = \max\limits_{\alpha: f \vdash g_\alpha} \operatorname{wt}(\alpha)$.

Proof: [Proof of Lemma 1] Without loss of generality consider $\alpha_0 = \alpha_1 = \ldots = \alpha_{r-1} = 1$, so $g_\alpha(x_0, x_1, \ldots, x_{m-1}) = x_0 x_1 \ldots x_{r-1}$. It is obvious that for

$$\varphi(x_0, x_1, \ldots, x_{m-1}) = \prod_{i=0}^{k-1} \left(\sum_{j=0}^{r-1} a_{i,j} x_j + b_i \right) \tag{11}$$

one has $\varphi \vdash g_\alpha$. Since the expression (11) for φ contains only r variables and $r = \operatorname{wt}(\alpha)$, $\varphi = g_\alpha + \rho$ where $\deg \rho < r$.

Now linear independency of the first r columns of $\mathbf{A}$ is to be proven. Consider it is not true and without loss of generality there is $l < r$ such that $\mathbf{A}^0 + \mathbf{A}^1 + \ldots + \mathbf{A}^l = 0$. One can transform x_j to $x_0 + x_j$ for every $j : 1 \le j \le l$. Then since $\deg \rho < r$ and the substitution is "linear" so it cannot increase degree of a polynomial,

$$\begin{aligned} &\varphi'(x_0, x_1, \ldots, x_{m-1}) \\ &= \varphi(x_0, x_1 + x_0, \ldots, x_l + x_0, x_{l+1}, \ldots, x_{m-1}) \\ &= x_0(x_1 + x_0)\ldots(x_l + x_0)x_{l+1}\ldots x_{r-1} \\ &\quad + \rho(x_0, x_1 + x_0, \ldots, x_l + x_0, x_{l+1}, \ldots, x_{m-1}) \\ &= g_\alpha(x_0, x_1, \ldots, x_{m-1}) + \rho'(x_0, x_1, \ldots, x_{m-1}), \end{aligned}$$

where $\deg \rho' < r$ as well. Thus, $\rho' \not\vdash g_\alpha$ and $\varphi' \vdash g_\alpha$.

On the other hand,

$$a_{i,0} x_0 + a_{i,j}(x_j + x_0) = (a_{i,0} + a_{i,j}) x_0 + a_{i,j} x_j,$$

so the substitution $x_j \mapsto x_j + x_0$ in (11) corresponds to replacing $\mathbf{A}^0 \mapsto \mathbf{A}^0 + \mathbf{A}^j$. Thus, φ' can be denoted by the expression (11) where the 0-th column of $\mathbf{A}$ is set to $\mathbf{A}^0 + \mathbf{A}^1 + \ldots + \mathbf{A}^l$ i.e. the all-zeros column. It means that x_0 is not present in the expression, because it has zero coefficient in every multiplier, so $\forall \beta \in \mathcal{I}_m$ $\beta_0 = 1$ implies $\varphi' \not\vdash g_\beta$. We came to the contradiction, because $\varphi' \vdash g_\alpha$. $\square$

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